

WRRRA_M 0.11 Sequential Calibration of the First Five Inputs

Wonsik Choi · 2026-10-02 · WRRRA Core 1.0 / MCC 2.3.2 · CC BY 4.0

Abstract

On fixed WRRRA Core 1.0 and MCC 2.3.2, WRRRA_M calibrates its first five physical inputs in the order $G \rightarrow H_0 \rightarrow f_{\text{phi}} \rightarrow f_{\text{c}} \rightarrow \text{electron rest energy}$. Each partial stage reads only supplied anchors and reports available outputs and remaining coefficients. The 0.10 energy-pressure calculation and nine physical cases are retained. Address-exponent and effective-admission reparameterization explicitly discloses additional arithmetic inputs. Adopting electron mode 23 and zero intercept gives a mass unit of 22,217.345682 eV. Two response shapes fitting the same anchors demonstrate reproduction together with residual freedoms.

Verification inputs

The complete 0.10 input is frozen by hash. G and electron energy use NIST 2022 CODATA; $H_0=67.4$ is the Planck 2018 base-LambdaCDM calibration benchmark. Standard uncertainties are 1.5×10^{-15} SI for G , 0.00016 eV for electron energy and $0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ for the adopted H_0 benchmark. Physical fractions 4.93% and 26.5% are inherited rounded calibrations without a covariance assigned here. The first five candidate-sheet entries and the MCC electron-mode relation are recorded as provenance.

Input	Adopted value	Unit / role
G	6.6743000e-11	$\text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
H_0	67.4	$\text{km s}^{-1} \text{ Mpc}^{-1}$
f_{phi}	0.0493	energy fraction
f_{c}	0.265	energy fraction
$m_{\text{e}} c^2$	510998.95069	eV

$c=299792458 \text{ m/s}$ is fixed. $h=6.62607015 \times 10^{-34} \text{ J s}$ and $1 \text{ eV}=1.602176634 \times 10^{-19} \text{ J}$ are exact SI references. The parsec conversion, local masses/radii/response function and conditional $\Phi=\Psi$ lensing are inherited. Address cutoff, four zeta heights, $K=8$, phase increment 0.1, response coefficients λ_{s} , volume exponents and the electron-mode integer are constitutive inputs; the five numerical anchors do not derive them.

WRRRA-specific transformation

The partial calculator adds one anchor at a time. G fixes the SI action normalization and curvature response. Exact h and c additionally give Planck length/time as unit references. Replacing all future anchors with NaN must leave the corresponding earlier-stage outputs unchanged.

$$C_G = \frac{c^4}{16\pi G}, B_G = \frac{8\pi G}{c^4}, \ell_P = \sqrt{\frac{\hbar G}{c^3}}, t_P = \ell_P / c. \quad (1)$$

After converting H0 to inverse seconds, the critical energy density is calculated. c/H0 and 1/H0 are Hubble length/time references, not assignments of the total cosmic size or age.

$$H_0 = \frac{H_{0, km/s/Mpc} 10^3}{10^6 pc_m}, u_{crit} = \frac{3 H_0^2 c^2}{8 \pi G}. \quad (2)$$

The phenotype fraction gives the total hidden load. Adding the clustering fraction f_c separates D from background R. eta_load is the homogeneous-carrier reference coefficient of 0.6, distinct from the address-sector eta_s coefficients of 0.10.

$$f_h = 1 - f_\varphi, f_b = 1 - f_\varphi - f_c, u_h = f_h u_{crit}, \eta_{load} = u_h / 2. \quad (3)$$

$$Q_h \equiv \zeta k_h^2 = \frac{16 \pi G u_h}{c^4}, L_* = Q_h^{-1/2}, L_0 = L_* \sqrt{W_0}. \quad (4)$$

The identifiable quantity is weighted twist Q_h. Stiffness zeta and twist kappa_h are not separately determined. L_star is a calculated length coefficient; L0 remains undetermined without W0. The 95.07% hidden share alone does not fix cosmic length.

Step	New input	Newly available outputs	Remaining anchors / choices
1	G	action coefficient; SI response	H0, f_phi, f_c, E_e
2	H0	H0 in s ⁻¹ ; critical density	f_phi, f_c, E_e
3	f_phi	hidden density; weighted twist	f_c, E_e
4	f_c	sector energies; a_T, P, q	E_e
5	E_e	mass unit; electron units	structural freedoms remain

The 0.10 address effects e_s(n), normalized weights w_n and positive responses g_s(n) are retained. Each eta_s is fitted once on the reference address state and uniform carrier, then held fixed for address/state controls. Carrier operators are A_phi=I, A_D=Kc/2 and A_R=Kb/2, each with unit uniform reference expectation.

$$\mu_s = \sum_n w_n e_s(n) g_s(n), \eta_s = \frac{f_s u_{crit}}{\mu_s}, \hat{E}(a) = V_0 \sum_s \eta_s \mu_s a^{\nu_s} A_s. \quad (5)$$

Pressure is the volume derivative of the same energy function. Retaining nu_phi=nu_D=0 and nu_R=3 gives pressureless D/phenotype and P_R=-u_R. Nonphenotype status does not delete energy or gravity.

$$V = V_0 a^3, E_s = \text{Tr}(\rho \hat{E}_s), P_s = -\partial E_s / \partial V = -\nu_s u_s / 3. \quad (6)$$

$$a_T = c H_0 \sqrt{f_c / 8}, P_0 = -f_b u_{crit}, q_0 = (1 - 3 f_b) / 2. \quad (7)$$

Electron mode n_e=23 is adopted in the zero-intercept MCC mass branch. The electron anchor fixes its common energy unit. Reported ±1, ±2, ±3 and ±23 are generic mode controls, not identifications of other particle masses. Mode sign has the same rest energy; charge remains separately inherited from the filter ledger. Mode number and zero intercept are additional constitutive choices.

$$E_n = \sqrt{E_{off}^2 + n^2 \mu_E^2}, E_{off} = 0, \mu_E = E_e / 23, E_e = m_e c^2. \quad (8)$$

The same total energy operator is expressed in electron-energy units. Multiplying back by E_e restores the original joule ledger. Electron energy is not separately added to cosmic density. Physical boundary conditions, the microscopic clock and general spectra remain for 0.12.

$$\tilde{E}_e = \hat{E} / E_e, \hat{E} = E_e \tilde{E}_e, m_e = E_e / c^2. \quad (9)$$

Re-expression of address alpha and beta

Here alpha_addr is the address power-law exponent and beta_eff is the effective admission rate of odd composites. These differ from electromagnetic fine-structure alpha_EM. The two independent 0.9 arithmetic targets D=0.268 and phi=0.05 remain disclosed additional inputs; physical energy targets 0.265 and 0.0493 do not replace them. C_even excludes the prime address 2.

$$w_n(\alpha_{addr}) = \frac{n^{-\alpha_{addr}}}{\sum_{j=2}^N j^{-\alpha_{addr}}}, \sum_{n \in C_{even}} w_n = 0.268. \quad (10)$$

$$\sum_{n \in C_{odd}} w_n b_n(h) = 0.05, \beta_{eff} = \frac{0.05}{\sum_{n \in C_{odd}} w_n}. \quad (11)$$

Resolving the fixed phase/filter gives alpha_addr=1.899687695055, h=1.447673170353 and beta_eff=0.865457012414. Alpha_addr and h fit the two arithmetic targets; beta_eff is then derived as their re-expression. A third independent beta input is unnecessary. The five physical anchors alone do not determine arithmetic targets or alpha_EM.

Calculated outputs

Quantity	Calculated value
ucrit J m ⁻³	7.668947767822e-10
u_hidden J m ⁻³	7.290868642868e-10
eta_load J m ⁻³	3.645434321434e-10
zeta kappa_h ² m ⁻²	3.028112744666e-52
Length coefficient m	5.746639841098e+25
a_T m s ⁻²	1.191812669106e-10
P Pa	-5.258597484395e-10
q0	-5.285500000000e-01
mu_E eV	2.221734568217e+04

The reference retains q0=-0.528550000, local rotation speed 207.510905127 km/s and conditional lens deflection 0.535586511 arcsec. The mass unit is 22217.345682174 eV. These are reproduction of validated benchmarks

and internal calculations under disclosed inputs. A new independent prediction is not a prerequisite for completion.

Residual freedoms are shown by calculated controls. Rescaling zeta and kappa_h as below preserves Q_h. Separately fitting eta_R at lambda_R=0 and 1 reproduces the same five targets and reference q, yet changing K to 4 after calibration gives different results. Fitting reference products eta_s mu_s and determining the full response shape are separate achievements.

$$\zeta \mapsto r \zeta, \kappa_h \mapsto \kappa_h / \sqrt{r}, \zeta \kappa_h^2 \mapsto Q_h. \quad (12)$$

lambda_R	Reference q	K=4 q after fitting
0.0	-0.528550000	-0.547819855
1.0	-0.528550000	-0.548643149

Independence of the five anchors is checked using the logarithmic sensitivity of $(C_G, u_{crit}, u_h, u_c, \mu_E)$. The numerical rank is five conditional on frozen constitutive choices. This is local identification of the five calibration anchors, not identification of every model coefficient.

$$\frac{\partial \log(C_G, u_{crit}, u_h, u_c, \mu_E)}{\partial \log(G, H_0, f_\phi, f_c, E_e)} = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 \\ -1 & 2 & -f_\phi/(1-f_\phi) & 0 & 0 \\ -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (13)$$

Ten runs changing each anchor by $\pm 1\%$ are construction-sensitivity controls, not observational confidence intervals. Changed physical targets are explicitly recalibrated; address controls hold SI coefficients fixed. Electron energy changes the mass unit but not macroscopic density, q or rotation speed. Critical density varies inversely with G and quadratically with H0.

The fixed-reference homogeneous expansion uses the same pressureless/background energy relation. Noncommuting-carrier internal exchange at fixed volume sums to zero and preserves state positivity/normalization. The required conversion work and its environmental account retain the 0.10 conservation checks.

$$\frac{H(a)^2}{H_0^2} = (f_\phi + f_c) a^{-3} + f_b, \frac{du}{d \log a} + 3(u + P) = 0. \quad (14)$$

Falsifiers and reproduction

The connection must be revised if an earlier stage reads future targets, input order changes, energy-pressure differentiation or inherited benchmark reproduction fails, electron energy is duplicated, or unidentified W0, zeta or alpha_EM is used as determined. This release executes 28 checks plus 32 inherited 0.10 checks, including zero/sign modes, five-point pressure differentiation, log sensitivities, explicit degeneracy controls, conservation and invalid-input rejection.

```
python -m pip install -r calculations/wrra_m_0_11/requirements.txt
python calculations/wrra_m_0_11/run_release.py
```

Parameters.json contains inputs, units, provenance and constitutive choices. Sequential calibration, physical cases, mass modes and input responses are supplied as JSON/CSV. A clean copy with all captured results

removed compares regenerated outputs and bilingual sources byte for byte. Word/PDF are final publications built from those sources, with equations/tables checked against execution and every rendered page inspected.

Input SHA256 67de28abf0741c406178a2fe4ce07962e69bce827c3196b78e887da3bd0dab13

Completion and subsequent scope

Version 0.11 completes staged calibration of the five disclosed anchors and reporting of residual freedoms. Their origins, a unique universe solution and calibration-free constants are not completion requirements. Version 0.12 treats clocks/boundaries/spectra, 0.13 physical quantization/post-observation states/records, and later stages uncertainty, stress, curvature, size and neutrino connections. Unmeasured outputs produced after fixing the model and calibration are recorded as WRRR predictions with their inputs and falsifiers.

References

Choi Wonsik. WRRR_M 0.10, frozen baseline commit 24707c512c3994f2b19a96d6f257ef7d1249dab3. <https://github.com/Wonsik-Choi-janefather/wrrr-m-0.1>

Choi Wonsik. Minimal Computation Cosmology 2.3.2, chapter 23, frozen commit 21daec110c0cbecb228c445d587eac8302e7f767. <https://github.com/Wonsik-Choi-janefather/minimal-computing-cosmology-2.3.2>

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Planck Collaboration. Planck 2018 results VI, A&A 641 A6 (2020). <https://doi.org/10.1051/0004-6361/201833910>

NIST. CODATA 2022 constants table. https://physics.nist.gov/cuu/pdf/wall_2022.pdf